TinyStar’Ex

The astronomical spectrograph you build yourself

Website: **tse-spectro.net**

TinyStar’Ex is a high-performance astronomical spectrograph, with its own built-in pointing system, that you build yourself. The complete optical kit is available from the shop; you print the mechanical parts and assemble them by following a detailed manual. It is simple, enjoyable and affordable. Compatible with standard deep-sky cameras, TinyStar’Ex suits both the curious beginner discovering spectroscopy and the experienced observer. And you learn a great deal by building and using it.

TECHNICAL SPECIFICATIONS

Grating	300 lines/mm, interchangeable for future applications
Slits	4 selectable widths: 14, 20, 26, and 32 µm
Optics	apochromatic doublets in low-dispersion glass
Focal lengths	collimator 50 mm, camera 50 mm
Aperture	telescopes as fast as f/4
Spectral range	375 to 840 nm in a single exposure (IMX585 sensor), extendable to 1,050 nm with an order-sorting filter (not supplied)
Resolution at 656 nm	0.84 nm (R = 780) with the 14 µm slit; 1.2 nm (R = 550) with the 20 µm slit
Main camera	M42 × 0.75, back focus 17.5 mm
Pointing / guiding	31.75 mm (1.25-inch) barrel
Assembly	design that minimizes adjustments
Mass	700 g without camera





RECOMMENDED CAMERAS

Two cameras are needed to get the most out of TinyStar’Ex.

- **A main camera**, for recording the spectra. Any model with a back focal length of 17.5 mm and an M42 × 0.75 (T2) interface will do. A cooled camera with an IMX585 sensor is ideal, thanks to its small pixels and its format. If you own another model, do not worry: it will at least let you take your first steps in astronomical spectrography, and very probably much more. Even a colour camera can be used. The first step is what counts!
- **A pointing and guiding camera**, to place a star or any other object on the slit and keep it there during the exposure. Conventional ‘stick’ guide cameras or entry-level planetary cameras are suitable.

THE CREATORS

STAROS Projects

TinyStar’Ex is designed by the **STAROS Projects** team, already behind the SUNSCAN spectroheliograph. It is a direct descendant of Star’Ex, conceived by Christian Buil.

RECOMMENDED TELESCOPES

- TinyStar’Ex performs at its best on telescopes of modest size, from 80 to 250 mm in diameter.
- It can be used on larger apertures, at the cost of a noticeable loss of spectral resolution, since the 26 or 32 µm slits must then be used. In return, the instrument becomes remarkably effective for applications where the signal-to-noise ratio matters more than resolution, such as measuring the spectral energy distribution (SED) or identifying very faint sources.
- In spectrography, reflecting telescopes are preferable to refractors. A mirror brings all wavelengths to the same focus, whereas a lens suffers from chromatic aberration, particularly in the blue. This is also true of so-called apochromatic refractors as soon as you move towards the ultraviolet. A refractor remains usable, at least for getting started, but TinyStar’Ex reveals its full power behind a reflector.
- A major strength of TinyStar’Ex is its ability to work with very fast telescopes, down to f/4, with virtually no loss of signal. Its preferred range is f/4 to f/8, but it remains versatile: with a Cassegrain or a Maksutov, for example, you will also obtain good spectra.